

SOC Home Lab Threat Emulation & Detection Verification Guide

1. Network Perimeter Telemetry (pfSense Firewall via Syslog)

Overview & Methodology

Evaluated perimeter visibility by configuring centralized Syslog forwarding from a pfSense 2.7.2 firewall to the Wazuh Manager over UDP port 514. Inbound port scan traffic was executed to validate stateful inspection logging and rule triggering on denied packets.

Execution Vector

From the external assessment instance (<ATTACKER_HOST>), a non-ping TCP SYN port scan was launched targeting common service ports on the firewall's external interface (<PFSENSE_WAN_IP>):

- `nmap -Pn -p 23,80,443,8080 <PFSENSE_WAN_IP>`

Forensic Evidence & SIEM Ingestion

The firewall's default drop policy intercepted the unauthorized ingress attempts, passing the raw CSV packet metadata (`filterlog`) to the SIEM.

Telemetry Field	Normalized Value	Technical Assessment
<code>rule.id</code>	100114	Wazuh custom firewall detection rule.
<code>rule.level</code>	7	Medium-high severity event indicating actively blocked ingress traffic.

rule.description	pfSense: Blocked traffic	Automated alert triggered on firewall discard events.
rule.groups	["pfsense", "firewall"]	Correlation taxonomies for SIEM alerting.
decoder.name	pfsense-custom	Custom regex/CSV decoder parsing FreeBSD filterlog structures.
full_log	filterlog[...]: ...,match,block,in,4,...	Validates directional packet block (in), Layer 4 protocol, and destination port telemetry.

```

PORT      STATE      SERVICE
23/tcp    filtered  telnet
80/tcp    filtered  http
443/tcp   filtered  https
8080/tcp   filtered  http-proxy

```

rule.description	rule.level	rule.id
pfSense: Blocked traffic	7	100114
pfSense: Blocked traffic	7	100114
pfSense: Blocked traffic	7	100114
pfSense: Blocked traffic	7	100114

2. Brute Force Authentication Attack (Hydra vs. SSH)

Overview & Methodology

Simulated an automated credential stuffing / dictionary attack targeting the OpenSSH daemon deployed on a Windows 11 endpoint to verify local logon auditing and threshold-based brute force alert generation.

Execution Vector

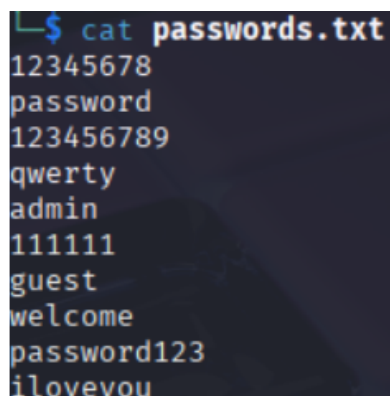
Using THC-Hydra, an automated password-guessing assault was launched using a structured dictionary against an administrative username:

- `hydra -l Administrator -P passwords.txt ssh://<TARGET_IP>`

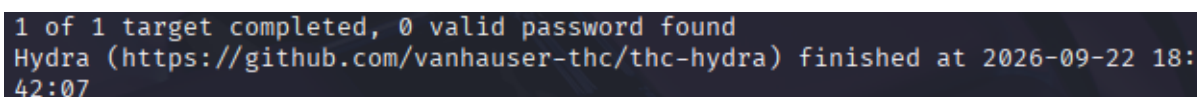
Forensic Evidence & SIEM Ingestion

Multiple sequential authentication rejections were recorded by the Windows Security Log (Event ID 4625), mapped via Sysmon/Security decoders, and correlated into alert streams:

- Initial Rejections ([rule.id: 60122](#), Level 5): *Logon Failure - Unknown user or bad password* triggered for each invalid credential attempt.
- Session Teardown ([rule.id: 67023](#) / [60137](#), Level 3): Recorded non-service account disconnection and logon failure logoffs.
- Post-Authentication / Success Tracking ([rule.id: 60106](#) / [67028](#)): Monitored successful logon and special privilege assignment events during baseline comparisons.



```
L$ cat passwords.txt
12345678
password
123456789
qwerty
admin
111111
guest
welcome
password123
iloveyou
```



```
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-22 18:
42:07
```

rule.description	rule.level	rule.id
Non service account logged off.	3	67023
Non service account logged off.	3	67023
Non service account logged off.	3	67023
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Logon Failure - Unknown user or bad password	5	60122
Windows User Logoff	3	60137
Special privileges assigned to new logon.	3	67028
Windows Logon Success	3	60106

3. Threat Intelligence & Endpoint Correlation (VirusTotal, AbuseIPDB, AlienVault OTX)

3.1. File Integrity Monitoring (FIM) & VirusTotal API Integration

- Mechanism: Tested real-time file creation monitoring using Wazuh Syscheck correlated with automated VirusTotal v3 API hash lookups.
- Payload Placed: EICAR Anti-Virus Test File placed inside a monitored path (C:\test\malware_test.txt).
- Detection (rule.id: 87105, Level 12):
 - Status: data.virustotal.malicious: 1
 - Detection Ratio: 66 security vendors/engines flagged the SHA-256 hash as malicious (data.virustotal.positives: 66).
 - Permalink Reference: Public VirusTotal analysis confirmed known signatures across industry AV engines.

rule.description	rule.level	rule.id
VirusTotal: Alert - c:\virustotaltest\malware_test.txt - 66 engines detected this file	12	87105
VirusTotal: Alert - c:\virustotaltest\malware_test.txt - 66 engines detected this file	12	87105

3.2. IP Reputation Correlation (AbuseIPDB Integration)

- Mechanism: Integrated external IP abuse verification triggered by failed authentication logs and anomalous external addresses.
- Detection ([rule.id: 100060](#), Level 7):
 - Alert Description: [AbuseIPDB: Suspicious IP detected - Score: 7% \(2 reports\)](#).
 - Enrichment: Automated geolocation resolution identifying source ASN, country, and abuse confidence score prior to analyst review.

rule.description	rule.level	rule.id
AbuseIPDB: IP sospechosa detectada (118.25.6.39) - Score: 7% (2 reportes)	7	100060

3.3. Threat Feed Validation (AlienVault OTX Integration)

- Mechanism: Queried AlienVault Open Threat Exchange pulses for known malicious scanners and reconnaissance nodes.
- Detection ([rule.id: 100070](#), Level 10):
 - Alert Description: [AlienVault OTX: Malicious IP detected - Associated with 22 pulse\(s\)](#).
 - Intelligence Detail: Identified correlated threat actor campaigns, auto-generated community pulses, and malicious scanner metadata.

rule.description	rule.level	rule.id
AlienVault OTX: IP maliciosa detectada (89.248.165.176) - Asociada a 22 pulso(s): jan2,2025 clone Auto-generate...	10	100070

4. Network & Host-Based Intrusion Detection (Suricata & Sysmon)

- Suricata NIDS ([rule.id: 86601](#), Level 3): Detected suspicious ICMP ping and transit sweep signatures across the monitored virtual switch interface.
- Sysmon Process Monitoring (Event ID 1 / [rule.id: 100051](#)): Flagged suspicious subshell and script interpreter spawning ([WindowsPowerShell](#)) during administrative tasks.
- Sysmon File Creation (Event ID 11 / [rule.id: 100052](#)): Monitored script and binary drops in temporal directories ([AppData\Local\Temp](#)).

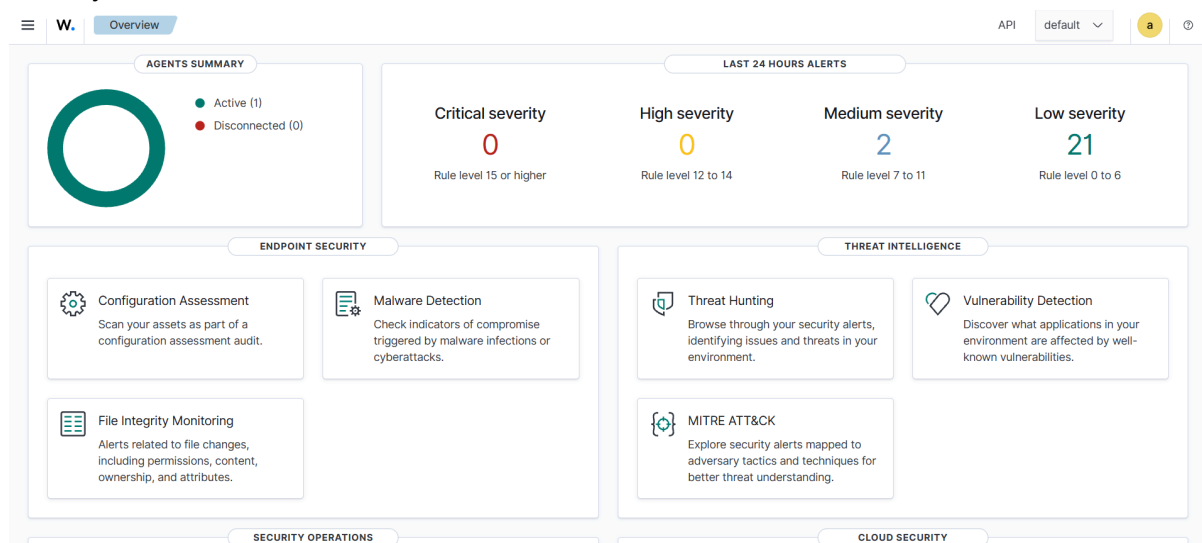
rule.description	rule.level	rule.id
Suricata: Alert - TEST: Ping detectado en la red	3	86601
Suricata: Alert - TEST: Ping detectado en la red	3	86601
Suricata: Alert - TEST: Ping detectado en la red	3	86601

Sysmon (ID 11): Creación de script o ejecutable en disco (C:\\Users\\rodril\\AppData\\Local\\Temp_PSScriptPo...	6	100052
Sysmon (ID 1): Ejecución de proceso sospechoso/consola (C:\\Windows\\SysWOW64\\WindowsPowerShell\\v1.0...	6	100051

5. MITRE ATT&CK Framework Mapping

MITRE ATT&CK Technique	ID	Detection Layer	Relevant Lab Event
Brute Force: Password Guessing	T1110.001	Host SIEM (Windows Security)	Hydra SSH attack generating Rule 60122.
Command and Scripting Interpreter: PowerShell	T1059.001	Endpoint Telemetry (Sysmon)	Sysmon Event 1 (rule.id: 100051) console invocation.
Malicious Ingestion / File User Execution	T1204.002	FIM + Threat Intelligence	Real-time file drop flagged by VirusTotal (Rule 87105).
Network Service Discovery	T1046	Perimeter Firewall (pfSense)	Inbound Nmap TCP SYN scan blocked by Rule 100114.
Network Traffic / Adversary Infrastructure	T1590	External Threat Intel	AbuseIPDB & AlienVault OTX alerts (100060, 100070).

Before (Pre-Attack Baseline): The SIEM overview establishes a clean baseline with only minimal background operational noise (23 low/medium events) and zero high or critical security alerts.



After (Post-Attack Telemetry): Following the attack simulations and threat intel integrations, total alerts surged past 1,540, registering 2 high-severity threat detections and 219 medium-severity authentication and firewall incidents.

